

Final Autonomous Capstone

Submission checklist, scoring rubric, and reflection

Student: _____

Date: _____

Pathway: Code Learning / Robot Programming

Team: _____

Challenge

Design, program, test, and explain a short autonomous routine using the VEXLearn template.

Minimum Requirements

- At least three movement commands
- At least one turn
- At least one mechanism action when hardware permits
- Planned starting position and route diagram
- Safe timeout or bounded movement behavior
- At least ten physical trials for the Robot Programming Path

Submission Checklist

Item	Included?	Location or notes
Route diagram		
Source code		
Hardware configuration		
PID values		
Trial log		
Demonstration video		
Failure analysis		
Final reflection		

Capstone Rubric

Category	Points	Score / feedback
Correct project and hardware configuration	15	
Understanding of C++ and project structure	15	
Driver-control or mechanism modification	10	
Autonomous planning and implementation	20	
PID tuning process	15	
Reliability testing and evidence	15	
Explanation, documentation, and reflection	10	
Total	100	

Performance Level

90-100 Independent / 75-89 Proficient / 60-74 Developing / Below 60 Incomplete

Final score: _____ / 100 Performance level: _____

Final Reflection

1. What part of the routine was least reliable?

2. Which change produced the largest improvement?

3. What evidence supports your final PID values?

4. What hardware behavior affected the software?

5. What would you change with another week?

6. When would JAR Template, LemLib, or another odometry-based framework become appropriate?
